NAZMUS SAKIB SAMI

Machine Learning Engineer | Software Engineer | Computer Science & Engineering Undergraduate

Dhaka, Bangladesh | +880 1310-334299 | saminazmussakib0@gmail.com

LinkedIn: linkedin.com/in/nazmussakibsami | GitHub: github.com/dnsakibss

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate at AIUB pursuing a career as a Machine Learning / AI Engineer. Hands-on experience building end-to-end machine learning pipelines in Python and scikit-learn, embedded IoT systems on ESP32, and backend applications in C#, .NET and SQL Server. Co-author of two peer-reviewed conference papers, including an IEEE ICCIT 2025 publication. Strong foundation in data structures, algorithms, object-oriented programming and database design; seeking an internship or entry-level role in AI/ML or software engineering.

EDUCATION

Bachelor of Science in Computer Science and Engineering — **Present**

American International University-Bangladesh (AIUB), Dhaka

Higher Secondary Certificate (HSC), Science | Ideal College, Dhanmondi — **2022**

Secondary School Certificate (SSC), Science | Narsingdi Model School — **2020**

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, Java, SQL, PHP

Machine Learning: scikit-learn, Supervised Learning, Classification, Random Forest, Support Vector Machine (SVM), Logistic Regression, Gradient Boosting, Model Evaluation (F1, ROC-AUC), Data Labeling, Data Leakage Prevention

Computer Science: Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Design, Role-Based Access Control (RBAC), Computer Graphics (OpenGL)

Frameworks: .NET Framework, ASP.NET MVC, ADO.NET, Windows Forms

Databases: Microsoft SQL Server, MySQL, MongoDB

Embedded Systems and IoT: ESP32, Arduino IDE, Sensor Integration (PPG, ECG, IMU), Signal Processing, Wi-Fi Telemetry, Cloud Integration

Tools and Platforms: Git, GitHub, Docker, Linux, Visual Studio, VS Code, Microsoft Office (Word, Excel, PowerPoint), Adobe Photoshop, Adobe Premiere Pro

Languages: Bangla (Native), English (Fluent)

PROJECTS

Wearable Fatigue Detection System | Python, scikit-learn, C++, ESP32 — **Capstone Project**

- Designed and built an ESP32 wearable integrating PPG (MAX30102), motion (MPU6050) and single-lead ECG (AD8232) sensors, with on-device R-peak detection and real-time LED and buzzer alerts.
- Streamed sensor telemetry over Wi-Fi to cloud services with optional Telegram notifications, and collected labeled ground-truth data using a physical push-button.
- Trained and compared Logistic Regression, Random Forest, HistGradientBoosting and SVM classifiers using an episode-level train/test split to prevent data leakage.
- Achieved 90.5% accuracy, 0.939 ROC-AUC and 0.786 F1-score with Random Forest on held-out episodes (single-subject, 35-minute dataset).

CivicLens 2.0: Civic Complaint Management Platform | C#, .NET Framework, Windows Forms, SQL Server

- Independently rebuilt and extended CivicLens v1 (a team project) into a full-featured complaint management application.
- Implemented role-based dashboards for five user roles (Citizen, Moderator, Police, Journalist, Admin) with an admin approval workflow for new registrations.
- Developed per-complaint real-time chat, media attachments, a paginated newsfeed with comments and a complete status-history audit trail.
- Engineered transactional complaint-assignment logic and admin tooling, including bulk user actions and safe cascade deletes.

Other Academic Projects

- **Hotel Management System** (Java): desktop application for managing hotel operations.
- **Library Management System** (PHP, MySQL): web-based system for managing library records and users.
- **4-Scene Transportation Environment** (C++, OpenGL): interactive four-scene computer graphics project.

PUBLICATIONS

- J. Ferdous, M. S. Zaman, **N. S. Sami**, K. W. Tabassum and M. M. Ahmmed, "Design and Development of Bio-Degradable IoT Systems for Sustainable Green Energy Monitoring and Management," 2025 28th International Conference on Computer and Information Technology (ICCIT), IEEE, 2025, pp. 1986-1991. DOI: 10.1109/ICCIT68739.2025.11491535
- K. W. Tabassum, **N. S. Sami**, J. Ferdous, M. S. Zaman and M. M. Ahmmed, "Cox's Bazar & the Blue Flag Standard: A Study on Coastal Tourism Sustainability," International Conference on Economics, Business and Technology Management (ICEBTM 2025), IUB, 2025, pp. 30-32. DOI: 10.67508/iub.icebtm.2025.004